Misurda Luca

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Portfolio | Biatorbágy, Hungary

Education

University of Óbuda — Budapest, Hungary
BSc in Computer Engineering, AI Specialization ***(GPA: 4.36/5.00)*** — *Sept. 2021 – July 2026 (ABD Status)*

Saint Margaret High School — Budapest, Hungary
Sept. 2017 – May 2021

Paul Ritsmann German Nationality Elementary School — Biatorbágy, Hungary
German Nationality Program — *2009 – 2017*

Certificates

High School Diploma — *May 2021*
- **Advanced Level Graduation Exams:** Informatics (Excellent), English Language (Excellent)
- **Result:** Excellent graduation rating in all subjects

Euroexam: B2 Complex English Language Exam — *March 2021*

ECL: B2 Written German Language Exam — *Dec. 2019*

ECL: B2 Oral German Language Exam — *April 2019*

Skills

Programming Languages: Python, C#, Java, JavaScript, TypeScript, SQL, Assembly, HTML5, CSS3

AI, Data & Machine Learning: OpenAI, Google Gemini, Ollama, HuggingFace, Mistral AI, **LM Studio**, LangChain, LlamaIndex, **LangFlow**, **EXO**, PyTorch, TensorFlow, Keras, Scikit-learn, NumPy, Pandas, Weights & Biases

Frameworks & Dev Tools: .NET, **React**, **Vite**, FastAPI, Flask, Unity, Streamlit, Git, Docker, Elasticsearch

Cloud & Enterprise Systems: AWS, Azure, ServiceNow, Dynatrace, Jira, SAP, TIMS, VOSIMA

Methodologies: Agile, Scrum, OOP, REST API, SDLC, Software Testing, Research Methodology, Vibe Coding

Experience

Vantage Towers Zrt. — Budapest, Hungary
AI Engineer Trainee (Data Analyst) & ServiceNow Architect Trainee (Dev) — *Sept. 2025 – Present*
- **AI Engineer Trainee (Data Analyst Team):** Development of complex **LLM and RAG architectures**, automated AI analysis of tower infrastructure plans, and support for data migration processes from **VOSIMA** systems to the **TIMS** database.
- **ServiceNow Architect Trainee (Development Team):** Design, customization, and development of enterprise-level platform architecture and complex process automation solutions within the ServiceNow ecosystem.
- **System Monitoring (Operations Team):** As the initial phase of the rotation program, stabilizing enterprise infrastructure, **Dynatrace** and **HPE OneView** based system monitoring, and supporting operational tasks.
- Daily professional cooperation in **English** with European peer teams following Agile **SCRUM** methodology.

József Attila High School Budapest, Hungary
Informatics / Digital Culture Adjunct Teacher *Sept. 2024 – July 2025*

- Theoretical and practical teaching of Informatics, **Python programming, and SQL** database management for 9th–12th-grade students.
- Successfully preparing students for **intermediate and advanced level graduation** exams, individual mentoring, and developing exam strategies.
- Conveying complex technical concepts in an accessible way and developing educational materials to deepen programming fundamentals.

Szilágyi Erzsébet High School Budapest, Hungary
Informatics / Digital Culture Adjunct Teacher *Sept. 2023 – July 2024*

- Teaching programming fundamentals (**C#, Python**) and **SQL** structures for **mathematics and informatics specialized** students (8th–11th grade).
- Methodological development of advanced curriculum, development of logical thinking and algorithmic skills in specialized classes.
- Talent management and competition preparation, supporting students during school and national IT competitions.
- Developing interactive lesson plans that bridge the gap between school curriculum and market software development expectations.

Awards & Professional Participation

Girls in Tech – Yettel Award: Winner of a technology competition with a unique **AI-based music analysis software**. The professional jury awarded a 1750 EUR scholarship for the innovative technical implementation. (Sept. 2025)

MedTech Challenge – Participant: Innovation competition organized by Óbuda University focused on the future of healthcare. Participating since June 2026, competing for an innovation grant of up to 50,000 EUR to develop scalable, AI-driven medical solutions. (June 2026 – Present)

GDE MIT Minds and Machines Hackathon – Participant: Innovation competition focused on developing artificial intelligence-based solutions for healthcare to increase the efficiency of diagnostics and data processing. (Feb. 2026)

Byborg AI Hackathon – Invited Participant: Selected based on professional screening for Byborg Enterprises' exclusive AI-focused developer competition, working on complex and creative multimedia challenges. (May 2024)

Projects

Systematic Examination and Evaluation of RAG Architectures – Au;ReliA | *GitHub*

- As part of my thesis project, I developed a **React** and **Vite** based, domain-agnostic **low-code platform**. Building on the **LangFlow** concept, it allows for the design and execution of state-of-the-art **RAG architectures** (e.g., **GraphRAG**, **Agentic RAG**) on an intuitive visual interface. I systematically examined and evaluated the effectiveness of various search strategies, measuring their **accuracy** and **relevance** while minimizing the risk of **hallucinations** through a built-in **multi-level validation layer** and source-critical nodes. The system's modular design supports the creation of custom workflows across industries, complemented by critical security features such as automated **PII masking** and robust, client-side **error handling**. The project combines scientific architectural analysis with practical applicability, providing a reliable and auditable tool for developing **enterprise-level AI solutions**.

IFC/CAD Data Processing Pipeline | *Vantage Towers – Non-public / Internal project*

- Implemented a robust data processing system for **Vantage Towers'** internal infrastructure project that transforms industry-standard **IFC (Industry Foundation Classes)** and **CAD models** into a format directly structurable and interpretable for AI (e.g., LLMs and RAG architectures). The solution automatically extracts, normalizes, and arranges into machine-processable semantic graphs complex **3D geometric features**, spatial topological relationships, and engineering metadata. The primary goal was to bridge the semantic gap between traditional engineering CAD software and modern **AI agents**, enabling **automated validation** and AI-based analysis of telecommunications tower and physical infrastructure design data.

Distributed Local LLM and RAG Architecture | *Vantage Towers – Non-public / Internal project*

- Expanded corporate AI infrastructure by implementing a client-server based, **local LLM** environment with a dedicated document management module and internal **RAG (Retrieval-Augmented Generation)** capabilities to support context-driven content generation. The system can collect supplementary data from the live internet while masking AI model external network traffic through a built-in **proxy layer** with routed anonymization, ensuring maximum enterprise data security. Local model testing and API serving were implemented using **LM**

Studio. To maximize computational capacity cost-effectively, I introduced the **EXO** framework to create a distributed inference network that **coordinates the resources of multiple Mac devices** in a cluster-like manner, thereby enabling the execution of resource-intensive models by optimizing existing workstations.

Enterprise Access Lifecycle Management | *Vantage Towers – Non-public / Internal project*

- Designed and implemented an enterprise-level, **ServiceNow-based access management framework** following "Privileged Access Management" (PAM) principles, which ensures full lifecycle management, transparent logging, and **audit-ready GRC compliance** with complex **Flow Designer** orchestration.
- Developed a dynamic **Access Extension** module with iterative "Wake and Check" logic, which eliminates entitlement creep through real-time recalculation and automated revision of permissions, while drastically reducing operational administrative burdens and security risks.

Minds-Machines Hackathon – Encephalon;Core | *GitHub*

- Developed a complex, **full-stack healthcare workflow prototype** that digitizes patient-side self-diagnosis and medical consultation processes on a single integrated platform. I created an interactive interface based on **React 19** and **Three.js** (**with a 3D pain map**), and backend services powered by **FastAPI**. The solution uses **Azure OpenAI** and **Google Gemini** models to automate the processing of consultation transcripts, generation of **SOAP documentation**, and performs **real-time drug safety checks** and **patient journey management** relying on an **Azure Cosmos DB** based data layer.

Byborg AI Hackathon – AI Automated Presentation Generator

- During an invitation-only professional competition, I collaborated with two teammates to develop a **generative AI tool** that automatically generates visually and content-wise structured **presentation slides** from unstructured input data. Using **large language models**, the software not only prepared the text but also created a unique **narrative** and **presenter style** suitable for the topic to automate workflows. The project demonstrated the practical efficiency of **generative artificial intelligence** in complex creative tasks, emphasizing the **business applicability** of the technology. As this was collaborative work, the source code is currently available in a private repository per teammates' decision, so it is not part of my public portfolio.

Corporate AI Agent Automation – Vantage Towers (TIMS)

- Collaborating with Portuguese and Hungarian intern colleagues, I developed an **AI agent-based automation system** for Vantage Towers' **TIMS platform**, which performed cleaning and updating of supplier data from heterogeneous sources. During the process, we extracted and validated information from **SAP**, **Vosima** (inherited from Vodafone), and **ServiceNow** databases using intelligent agents, significantly increasing **data accuracy** and **process efficiency**. We automated complex **corporate data migration** in an international environment, paying close attention to strict **data security requirements**. Due to sensitive business data and corporate non-disclosure (**NDA**), the source code is not public, but the experience gained is relevant to any **large-enterprise AI integration**.

Animal Disease Classifier System | *GitHub*

- Developed a **Python-based machine learning application** capable of classifying animal diseases into **dangerous** or **non-dangerous** categories based on provided symptoms. Used a dataset from **Kaggle**, including raw data cleaning, handling missing values, and **feature engineering**. Built an interactive **Streamlit-based** user interface around the trained **classifier model**, allowing for **real-time interaction** with the model and immediate **visualization** of predictions.

MRI-based Brain Tumor Classifier System | *GitHub*

- Developed **machine learning models** for the automated classification of three different types of **brain tumors** visible on **MRI scans**. Applied complex **image processing techniques** (such as **Canny edge detection**, **CLAHE contrast enhancement**, as well as **HOG** and **SIFT** feature extraction) to prepare the data. Conducted a detailed **comparative analysis** of the efficiency of different methods and evaluated the performance of models trained on original and **PCA (Principal Component Analysis)** reduced feature sets, optimizing **recognition accuracy**.

Music Genre Classifier and Creative Support System | *GitHub*

- Developed a **machine learning system** that predicts the **three most likely musical genres** based on relevant **audio features**, organizing thousands of unique genres from **Spotify** into manually structured main categories. During development, I placed great emphasis on the **mathematical validation** of the models and a **detailed evaluation** of their performance, for which I also prepared a scientific-standard **conference publication draft**. I am currently developing the solution into an **interactive web platform** aimed at **supporting musicians' creative workflows** using artificial intelligence.

Game of Thrones Themed Digital Chess Application | *GitHub*

- Designed and developed a custom desktop chess application in **C#** using **object-oriented programming** principles and the **WPF (Windows Presentation Foundation)** framework. A unique feature of the project is the self-made, **Game of Thrones-inspired pixel art visual world**, where every piece and interface element fits the theme. Beyond **implementing classic chess logic**, I focused on **user experience** and **responsive graphical rendering**, combining traditional gameplay with a modern, atmospheric design.

Corporate Network Infrastructure Design and Simulation | *GitHub*

- Designed and simulated the IT ecosystem of a complex clothing store, covering the entire network architecture and server-side infrastructure. I built the network using **Cisco Packet Tracer**, configured a **Windows Server 2022** based domain controller in a **VirtualBox** environment, and implemented a HTML-based website. I gained in-depth experience in IP addressing, security policies, and enterprise service management.

Corporate Procurement Process Modeling and SQL Data Management

- Designed the **strategic procurement processes** of a fictional electronics manufacturing company, modeling and visualizing the workflows in **Signavio**. I designed an **SQL database structure** for the **efficient management** of supplier data, orders, and inventory. The task helped in understanding the synergy between **business logic** and **IT backend systems** and deepening **process optimization skills**.

Recipe Management CRUD Application and Testing Framework | *GitHub*

- Developed a **console-based recipe management system** in **C#** featuring full **CRUD** (Create, Read, Update, Delete) functionality and **SQL** database management. I placed special emphasis on **software quality**: I ensured code **robustness** and minimized logical errors by writing comprehensive **Unit tests**, demonstrating the application of a **test-driven approach** in software development.

Relevant University Coursework

Professional Foundation Studies: Analysis I-II, Discrete Mathematics and Linear Algebra I-II, Introduction to Informatics, Physics, Electrical Engineering, Microeconomics, Macroeconomics, Software Design and Development I-II, Electronics, Computer Networks, Business Economics I-II, Probability and Mathematical Statistics, Fundamentals of Management, Public Administration and Legal Studies, Modern Computer Architectures I-II, Web Programming and Advanced Development Techniques, Fundamentals of Computer Architectures, Project Work I-IV, Databases, Infocommunication Techniques, Enterprise Information Systems, Thesis I-II, Systems Theory, Software Technology and GUI Design, Digital Systems, Intelligent Systems, Operating Systems, IT Security, Professional Comprehensive Exam

Specialization Studies: Introduction to Data Science, Machine Vision and Image Processing in Healthcare, Deep Machine Learning, Big Data and Cloud-based Services, Advanced Robot Programming in ROS Environment

Future Goals

At the intersection of **artificial intelligence**, **software development**, and **enterprise systems**, I want to develop practical solutions that create real business and social value. To further my professional growth, I plan to obtain a **C1-level English language certificate** and actively participate in **foreign professional programs** and **research projects** to expand my knowledge in an international environment.

In the long term, I aim to become a professional who is not only strong in designing modern **AI-based architectures** but is also capable of shaping them into **reliable** and **scalable** systems, whether it concerns **RAG solutions** or **data-driven automation**.

I am also passionate about **teaching**, so my goal is to leverage my experience in an instructional or mentorship role, contributing to the education of future professionals.

Hobbies

Cooking / Baking | Hiking (Blue Trail) | Gardening | Video Games | Comic Books / Manga | Content Creation / Video Editing